

Open X-Embodiment · arXiv 2310.08864 · Nov 2 · Data

Scientific Reviewer

8+3

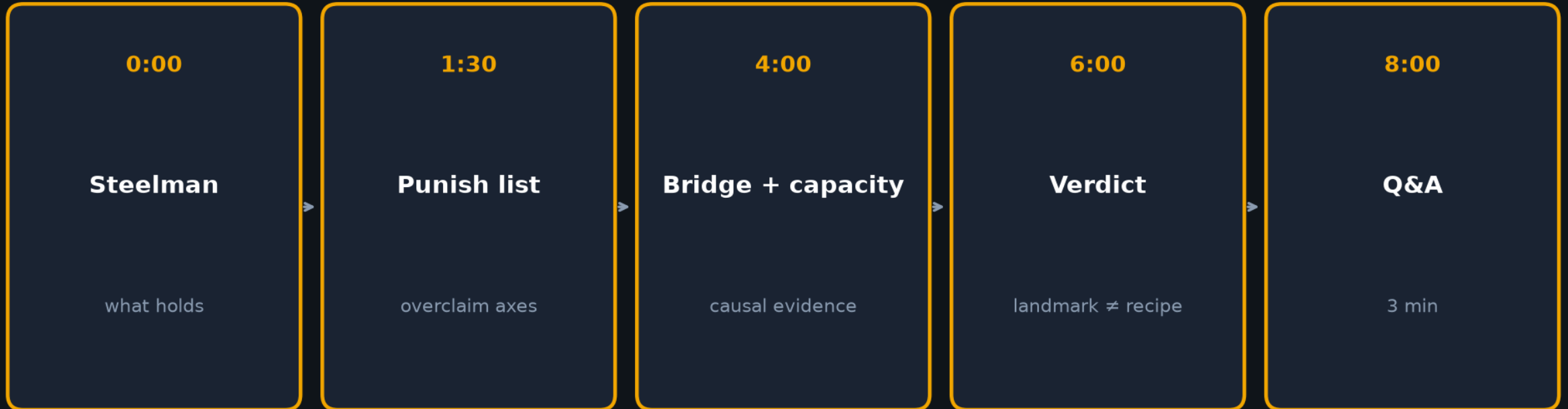
Landmark systems+dataset · punish overclaim
9<<22 · not new-robot ZS · not BEHAVIOR recipe

Paper club · Data cluster · due 9:30 AM PT

Spine: pool $\neq$ train mix · capacity $\times$ mix · steal mix science, don't drink the river

Open X-Embodiment Collaboration · 21 institutions · robotics-transformer-x.github.io

Scientific Reviewer talk arc · 8+3



Spine: pool ≠ train mix · capacity×mix · steal mix science, don't drink the river

Scientific Reviewer — holds vs punish

HOLDS

- ✓ Matched RT-1 vs RT-1-X isolates mix on small domains
- ✓ Capacity caveat on large domains is honest (Table I)
- ✓ Bridge ablation = causal-ish evidence for transfer
- ✓ Web / history / scale axes (Table II) are right
- ✓ Limitations flag new-robot + sensor-gap correctly

PUNISH / QUALIFY

- × “Generalist X-robot” $\neq$ zero-shot new hardware
- × Train mix 9 $\ll$ pool 22 — freeze mix cards
- × Heterogeneous abs/rel/vel $\neq$ universal interface
- × PaLM skill ontology = figure aid, not taxonomy
- × 55B RT-2-X unreproducible; lean on open successors
- × No crisp negative-transfer decision rule

CLEAR: Pool ≠ RT-X train mix

Marketing blur is the #1 citation failure mode

OXE POOL

Datasets	60
Embodiments	22
Institutions / labs	21 / 34
Trajectories	1M+
Skills / tasks	527 / 160,266
Scenes	311
Schema	RLDS TFRecords

RT-X TRAIN MIX

Manipulators	9 (≠ full 22)
Sources (ex.)	RT-1, QT-Opt, Bridge, Play, Jaco, Cable, ...
RT-1-X	robotics mix only
RT-2-X	~1:1 web : robotics
Eval	3600 trials / 6 robots
Downstream	Octo ~800k · OpenVLA ~970k

Moral: every claim needs a frozen mix card — pool size ≠ train size ≠ Octo/OpenVLA subset

Cite “22 robots / 1M+” for the commons; cite “9 manipulators” for RT-X experiment results

Capacity × mix co-design (Table I moral)

More data is not free — small models underfit rich X-mixes on large native domains

RT-1-X (35M) on X-mix

Google 6-skill: 73% (vs RT-1 92%)

Bridge IRIS: 27% (vs RT-1 40%)

Underfits large native sets

RT-2-X (55B) on X-mix

Google 6-skill: 91% (recovers)

Bridge IRIS: 50% (beats)

Capacity absorbs diversity

BEHAVIOR implication

- If shipping $\pi_{0.5}$ / GROOT-scale → you can afford OXE-like width (with weight table).
- If shipping a tiny policy → curate narrower; gate external shards behind validation q-score.
- Empiricist underfit probe: same heavy mix, small vs base open ckpt, matched steps.

Table II — Bridge ablation (emergent skills)

Bridge→Google Robot skills absent from Google-only data

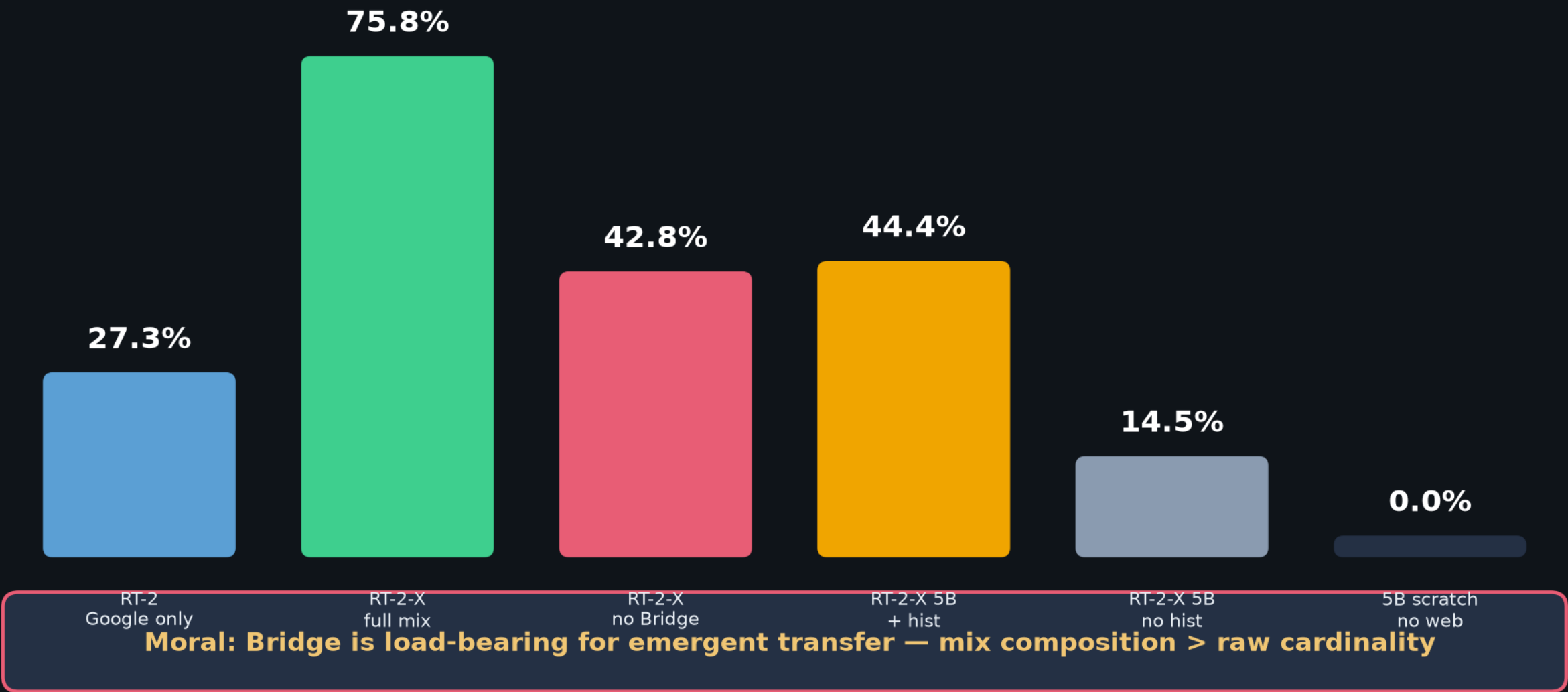


Table II — Emergent skills / Bridge ablation / capacity

(1)	RT-2	55B	none	Google Robot action	Yes	Web-pretrained	27.3%	62%
(2)	RT-2-X	55B	none	Robotics data	Yes	Web-pretrained	75.8%	61%
(3)	RT-2-X	55B	none	Robotics data except Bridge	Yes	Web-pretrained	42.8%	54%
(4)	RT-2-X	5B	2	Robotics data	Yes	Web-pretrained	44.4%	52%
(5)	RT-2-X	5B	none	Robotics data	Yes	Web-pretrained	14.5%	30%
(6)	RT-2-X	5B	2	Robotics data	No	From scratch	0%	1%
(7)	RT-2-X	5B	2	Robotics data	No	Web-pretrained	48.7%	47%

TABLE II: Ablations to show the impact of design decisions on generalization (to unseen objects, backgrounds, and environments) and emergent skills (skills from other datasets on the Google Robot), showing the importance of Web-pretraining, model size, and history.

B. Improved generalization to out-of-distribution settings

We now examine how X-embodiment training can enable better generalization to out-of-distribution settings and more complex and novel instructions. These experiments focus on the high-data domains, and use the RT-2-X model.

Unseen objects, backgrounds and environments. We first conduct the same evaluation of generalization properties as proposed in [9], testing for the ability to manipulate unseen objects in unseen environments and against unseen backgrounds. We find that RT-2 and RT-2-X perform roughly on par (Table II, rows (1) and (2), last column). This is not unexpected, since RT-2 already generalizes well (see [9]) along these dimensions due to its VLM backbone.

Emergent skills evaluation. To investigate the transfer of knowledge across robots, we conduct experiments with the Google Robot, assessing the performance on tasks like the ones shown in Fig. 5. These tasks involve objects and

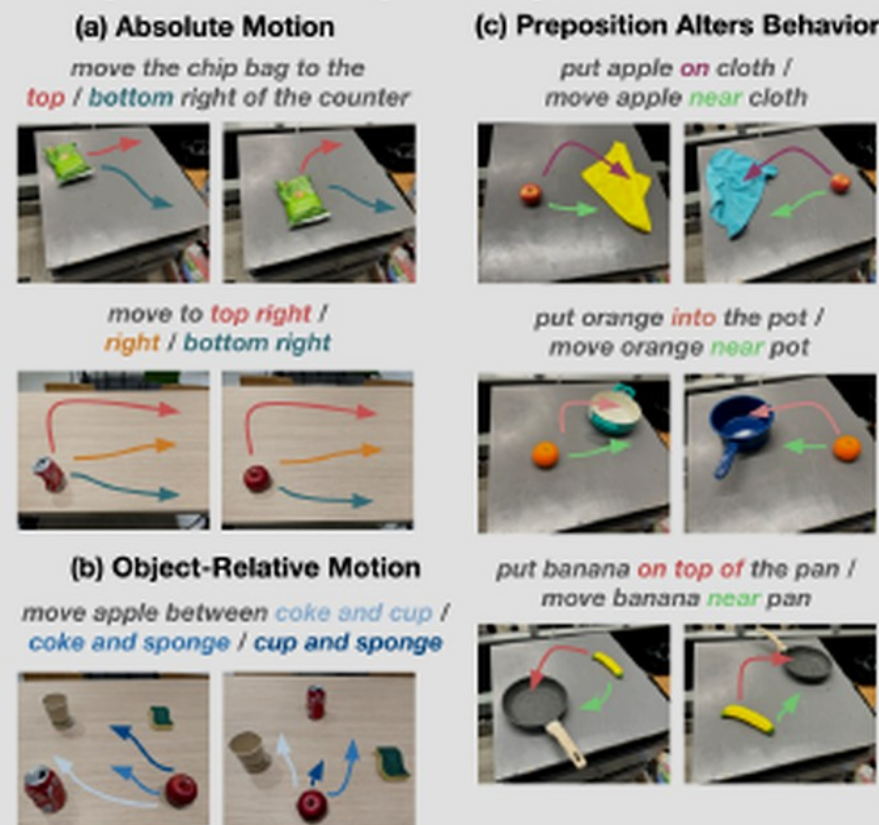


Fig. 5: To assess transfer between embodiments, we evaluate the

strong performance in these two evaluation scenarios.

Table [91] datasets. RT-1-X is trained on only robotics mixture data defined above, whereas RT-2-X is trained via co-fine-tuning (similarly to the original RT-2 [9]), with an approximately one to one split of the original VLM data and the robotics data mixture. Note that the robotics data mixture used in our experiments includes 9 embodiments which is fewer than the entire Open X-Embodiment dataset (22) – the practical reason for this difference is that we have continued to extend the dataset over time, and at the time of the experiments, the dataset above represented all of the data. In the future, we plan to continue training policies on the extended versions of the dataset as well as continue to grow the dataset together with the robot learning community.

At inference time, each model is run at the rate required for the robot (3-10 Hz), with RT-1 run locally and RT-2 hosted on a cloud service and queried over the network.

V. EXPERIMENTAL RESULTS

Our experiments answer three questions about the effect of X-embodiment training: (1) Can policies trained on our

tion and robot embodiment as in the respective publications. For large-scale dataset experiments, we consider Bridge [95] and RT-1 [8] for in-distribution evaluation and use their respective robots: WidowX and Google Robot.

For each small dataset domain, we compare the performance of the RT-1-X model, and for each large dataset we consider both the RT-1-X and RT-2-X models. For all experiments, the models are co-trained on the full X-embodiment dataset. Throughout this evaluation we compare with two baseline models: (1) The model developed by the creators of the dataset trained only on that respective dataset. This constitutes a reasonable baseline insofar as it can be expected that the model has been optimized to work well with the associated data; we refer to this baseline model as the *Original Method* model. (2) An RT-1 model trained on the dataset in isolation; this baseline allows us to assess whether the RT-X model architectures have enough capacity to represent policies for multiple different robot platforms simultaneously, and whether co-training on multi-embodiment data leads to higher performance.

Headline numbers (tattoo)

Fig. 4

~+50%

relative mean SR lift
small-data domains
RT-1-X vs Original / RT-1

Table I

73% vs 92%

RT-1-X underfits
Google Robot 6 skills
capacity × mix

Table II

27.3 → 75.8

emergent skills ~3×
Bridge→Google Robot
X-mix transfer

Ablation

42.8%

no-Bridge mix
emergent collapses
Bridge is load-bearing

Table I — large native domains (exact)

Setting	Original	RT-1	RT-1-X	RT-2-X 55B
Bridge IRIS WidowX	13%	40%	27%	50%
Bridge RAIL WidowX	13%	30%	27%	30%
Google 6 skills	—	92%	73%	91%

RT-1-X hurts large in-domain SR — positive transfer is conditional on capacity

Landmark systems + dataset paper

Positive transfer under X-embodiment co-training holds.
Capacity needed for rich mixes holds.

**Do NOT inflate to “solved cross-embodiment FM”
or “2026 BEHAVIOR data recipe complete.”**

Cite: shared RLDS pool + RT-X existence proof + Bridge ablation discipline.

Prefer open successors (Octo / OpenVLA / π_0) for reproducible science.

Scientific Reviewer · close

Landmark systems+dataset paper

Not complete BEHAVIOR recipe · not new-robot ZS

Holds: matched RT-1 vs RT-1-X · capacity caveat · Bridge ablation
Punish: 9<<22 blur · generalist marketing · 55B unrepro

Q&A · keep the spine clear

Open X-Embodiment

Robotic Learning Datasets and RT-X Models

arXiv 2310.08864 · Nov 2 Data cluster

OXE = municipal water · BEHAVIOR'26 = bottled specialty blend

Steal mix science — don't drink the river

Pool: 60 ds / 22 emb / 1M+ traj Train mix: 9 manipulators Eval: 3600 trials / 6 robots

Headlines: Fig.4 ~+50% · Table I RT-1-X 73% vs RT-1 92% · emergent 27.3%→75.8% · no-Bridge 42.8%

Seat 1 Visionary = Kanta Saito · Quiet build